

BRIER ISLAND, NS, Canada

WMO: 719880

Lat: **44.286N**Lon: **66.343W**Elev: **16**StdP: **101.14**Time Zone: **-4.00 (NAA)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-10.1	-14.9	1.0	-10.2	-13.7	1.2	-8.9	21.4	-3.1	19.3	-3.1	11.6	300	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	21.4	18.2	20.0	17.5	18.9	16.8	19.1	20.5	18.1	19.4	17.2	18.4	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	19.8	17.5	12.6	18.8	16.7	11.9	18.0	54.3	20.6	51.2	19.4	48.5	18.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.1	14.2	-14.4	24.6	2.4	1.6	-16.1	25.7	-17.5	26.7	-18.9	27.6	-20.6	28.8		
(5)			-14.7	21.1	2.2	1.2	-16.3	21.9	-17.5	22.6	-18.8	23.3	-20.3	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-1.2	-1.0	1.4	5.3	9.1	12.6	15.5	16.1	14.5	11.1	6.6	2.1	
	DBStd	6.85	4.80	4.19	3.50	2.45	2.09	2.12	1.80	1.62	2.02	2.68	3.68	4.20	
	HDD10.0	1490	348	309	268	141	43	3	0	0	0	20	111	247	
	HDD18.3	3883	606	543	526	390	286	173	89	73	117	224	352	504	
	CDD10.0	653	0	0	0	1	16	81	171	188	134	54	9	1	
	CDD18.3	4	0	0	0	0	0	0	2	2	1	0	0	0	
	CDH23.3	6	0	0	0	0	0	0	3	2	2	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	6.6	8.9	8.6	7.9	6.6	5.3	4.8	4.5	4.2	4.8	6.8	8.0	8.7	
Precipitation	PrecAvg	1260	112	96	107	95	94	86	84	78	107	121	124	138	
	PrecMax	1694	210	192	217	178	203	263	167	182	262	297	222	265	
	PrecMin	839	44	37	19	13	28	30	23	27	56	41	49	53	
	PrecStd	184	36	36	44	41	45	51	42	38	53	56	44	48	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	9.0	11.1	13.2	17.6	20.8	23.3	23.1	22.0	18.8	16.2	12.7	
		MCWB	10.0	8.2	9.1	10.3	13.1	16.6	19.3	19.3	19.0	17.1	15.0	12.1	
	2%	DB	8.8	7.0	8.6	11.0	15.0	18.6	21.2	21.3	20.0	17.0	14.4	11.0	
		MCWB	8.2	6.2	7.2	8.7	11.7	15.5	17.9	18.5	17.8	15.7	13.5	10.6	
	5%	DB	7.2	5.7	7.1	9.7	13.5	17.0	19.7	20.0	18.5	15.8	12.9	9.4	
		MCWB	6.4	4.8	5.9	7.8	11.1	14.6	17.2	17.8	16.8	14.5	12.0	8.6	
	10%	DB	5.5	4.5	5.7	8.5	12.2	15.8	18.4	18.8	17.4	14.8	11.7	7.9	
		MCWB	4.4	3.6	4.5	6.9	10.2	13.6	16.5	17.0	16.0	13.6	10.5	6.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	8.4	9.7	11.9	14.4	17.8	20.1	20.6	19.7	17.5	15.4	12.3
			MCDB	10.6	8.7	10.7	12.5	16.3	19.7	22.0	22.0	21.0	18.3	15.9	12.6
2%		WB	8.2	6.5	7.5	9.5	12.7	16.3	18.7	19.1	18.2	16.0	13.7	10.7	
		MCDB	8.7	6.9	8.3	10.3	14.1	17.9	20.4	20.5	19.4	16.7	14.2	11.0	
5%		WB	6.5	5.0	6.1	8.2	11.5	15.1	17.6	18.2	17.2	14.8	12.2	8.9	
		MCDB	7.1	5.6	7.0	9.2	12.9	16.4	19.1	19.4	18.1	15.6	12.8	9.3	
10%		WB	4.5	3.7	4.7	7.2	10.5	14.0	16.8	17.3	16.2	13.8	10.7	7.1	
		MCDB	5.3	4.4	5.5	8.3	11.9	15.2	18.0	18.5	17.2	14.7	11.5	7.8	
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.4	4.8	4.7	5.4	6.1	6.3	5.9	4.9	4.2	4.5	5.1	
		MCDBR	8.3	6.5	6.6	6.3	7.4	8.1	8.2	7.5	6.5	5.4	5.7	7.0	
		MCWBR	8.7	6.5	5.9	4.6	4.9	5.4	5.5	5.2	4.8	4.6	5.9	7.6	
	5% WB	MCDBR	8.3	6.6	6.8	5.9	6.8	7.5	7.5	6.9	6.1	5.2	5.9	7.0	
		MCWBR	8.9	6.9	6.5	5.0	5.2	5.7	5.7	5.4	5.0	5.0	6.4	7.6	
Clear-Sky Solar Irradiance	taub	0.306	0.308	0.333	0.355	0.378	0.408	0.424	0.405	0.368	0.341	0.324	0.308		
	taud	2.479	2.498	2.454	2.429	2.411	2.342	2.327	2.379	2.444	2.518	2.523	2.491		
	Ebn at Noon	821	889	906	909	893	863	844	849	856	837	794	781		
	Edh at Noon	70	82	98	109	114	123	123	113	98	80	68	64		
All-Sky Solar Radiation	RadAvg	1.28	2.20	3.50	4.48	5.09	5.63	5.65	5.10	4.18	2.69	1.52	1.05		
	RadStd	0.11	0.20	0.26	0.43	0.63	0.54	0.45	0.32	0.29	0.28	0.17	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	+0.49	N/A	N/A	N/A	+69	+1
(47) Regional (23 neighbors)		N/A	N/A	+1.04	N/A	+0.41	+0.51	N/A	N/A	N/A

Nomenclature: See separate page